

KCPC Week 1 Solutions

1 Logic

Basic – Same Birthday Month

There are only 12 months. If 13 students are placed into 12 month-boxes by birth month, the pigeonhole principle forces at least one box to contain at least two students.

Easy – Glove Selection

For one matching pair, the adversarial draw is all gloves of the same hand: $5 + 3 + 2 = 10$ gloves. The next glove guarantees a pair, so the answer is 11.

For one matching pair of each colour, the worst case is all 10 black gloves, all 6 brown gloves, and 2 grey gloves of the same hand. The next grey glove completes the final pair, so the answer is 19.

Medium – Three People: Knights, Knaves, and Spies

Let the types be K (Knight), N (Knave), S (Spy). Exactly one of each exists. Alice says “Bob is N ”; Bob says “Carol is K ”; Carol says “Alice is S ”.

Suppose Alice is K . Then Bob is N , so Bob lies. Bob says “Carol is K ”, so Carol is not K . Carol must be S . Then Alice is K , Bob is N , Carol is S – consistent. Carol says “Alice is S ”, which is false; as Spy she may lie. This works.

Suppose Bob is K . Then Carol is K , contradiction (two Knights).

Suppose Carol is K . Then Alice is S , so Alice is not K . Bob must be K (only one K), but then Carol is K – contradiction.

The only consistent assignment: Alice = Knight, Bob = Knave, Carol = Spy.

Key insight: Test each possible assignment of the unique Knight; contradictions narrow it down.

Hard – Islander Statements

Let L be the number of liars. Statement i is true exactly when $L \geq i$. Therefore the true statements are precisely $1, 2, \dots, L$, so there are L truth-tellers and $10 - L$ liars. But L is the number of liars, so $L = 10 - L$, hence $L = 5$. Islanders 1 through 5 tell the truth; islanders 6 through 10 lie.

Very Hard – Counterfeit Coin

One standard strategy is as follows. Label the coins 1 to 12. First weigh 1, 2, 3, 4 against 5, 6, 7, 8.

If they balance, the counterfeit is among 9, 10, 11, 12. Weigh 9, 10, 11 against 1, 2, 3. If this balances, coin 12 is counterfeit; compare it with coin 1 to decide heavy/light. If it does not balance, the odd coin is one of 9, 10, 11 and its direction is known; weigh 9 against 10 to identify it.

If the first weighing is unbalanced, suppose 1, 2, 3, 4 are heavier than 5, 6, 7, 8; the other case is symmetric. Then the counterfeit is either heavy among 1, 2, 3, 4 or light among 5, 6, 7, 8. Weigh 1, 2, 5 against 3, 6, 9. If left is heavy, the counterfeit is heavy 1 or 2, or light 6; weigh 1 against 2 to decide. If right is heavy, the counterfeit is heavy 3 or light 5; weigh 3 against 9 to decide. If balanced, the counterfeit is heavy 4, light 7, or light 8; weigh 7 against 8 to decide.

2 Invariants

Basic – Jigsaw Assembly

Each move reduces the number of sections by exactly 1. Starting with 500 sections and ending with 1 requires 499 moves.

Easy – Odd Handshakes

Sum the number of handshakes made by each person. This total equals twice the number of handshakes, so it is even. The people with even handshake counts contribute an even amount to the sum. Therefore the people with odd handshake counts must also contribute an even amount. A sum of odd numbers is even only when there are an even number of them.

Medium – Chameleons

Track the differences between colour counts modulo 3. A meeting changes two colour counts by -1 and the third by $+2$, so differences are unchanged modulo 3. Initially the differences include $14 - 10 = 4 \equiv 1$, $15 - 14 = 1 \equiv 1$, and $15 - 10 = 5 \equiv 2 \pmod{3}$. If all colours became equal except one colour present, some relevant difference would be $0 \pmod{3}$. The invariant prevents this, so it is impossible.

Hard – The MU Puzzle

Track the number of I's modulo 3. Initially it is 1. Rules 1 and 4 do not change it, Rule 2 doubles it, and Rule 3 subtracts 3. Starting from a number not divisible by 3, these operations never make it divisible by 3. The string MU has 0 I's, divisible by 3, so it is unreachable.

Very Hard – The Fifteen Puzzle

Associate a board with the permutation of tiles read row by row. Legal horizontal moves preserve inversion parity; legal vertical moves change inversion parity and also change the blank row parity. The combined invariant is preserved. Swapping 14 and 15 changes inversion parity but leaves the blank position unchanged, so the target is unreachable.

3 Colouring Arguments

Basic – One Missing Corner

No. A domino covers two squares, so any domino tiling covers an even number of squares. The board has 63 squares.

Easy – Corner-To-Corner Knight Journey

No. A knight alternates square colour every move. Visiting all 64 squares uses 63 moves, an odd number, so the start and end squares would need opposite colours. Opposite corners of a chessboard have the same colour.

Medium – Board Walks

Colour the shapes like a chessboard. A path using horizontal/vertical moves alternates colours. If the counts differ by more than 1, no Hamilton path exists. For the first cross, the colour imbalance rules out a path. For the second, the colour counts differ by 1, and a path can be drawn by snaking through the central rows and then through the arms; instructors can sketch this live.

Hard – Jumping To The Other Side

Colour the board as a chessboard. Every legal jump lands on a square of the same colour as the starting square. Initially there are 9 counters on one colour above the diagonal, but below the diagonal there are only 6 target squares of that colour. Since colour occupancy is invariant, the task is impossible.

Very Hard – Domino Tiling Revisited

The answer is: exactly even n . Odd n leaves an odd number of remaining squares after removing two squares, so tiling is impossible. For even n , a constructive argument using corridor paths shows that any two opposite-colour missing squares can be joined and the remaining board tiled by dominoes.

4 Induction And Monovariants

Basic – Polygon Triangulation

Base case $n = 3$: a triangle is already triangulated, with $3 - 2 = 1$ triangle. Assume any convex k -gon triangulates into $k - 2$ triangles for $3 \leq k < n$. For an n -gon, pick any diagonal that splits it into a k -gon and an $(n - k + 2)$ -gon with $3 \leq k \leq n - 1$. By

induction, the triangulation has $(k - 2) + ((n - k + 2) - 2) + 1 = n - 2$ triangles (the $+1$ is the triangle formed by the diagonal and the two vertices). All triangulations have the same count.

Key insight: Induction on the number of sides; any diagonal reduces the problem to smaller polygons.

Easy – Induction Inequality

Base case: $4! = 24 > 16 = 2^4$. Assume $n! > 2^n$ for some $n \geq 4$. Then

$$(n + 1)! = (n + 1)n! > (n + 1)2^n > 2 \cdot 2^n = 2^{n+1}.$$

Medium – Infected Chessboard

The answer is n . Sufficiency: infect one full diagonal; infection spreads outward until the board is infected. Necessity: the perimeter of the infected region never increases when a new square is infected, because at least two boundary edges are absorbed and at most two are added. Fewer than n starting squares have total perimeter less than $4n$, so they cannot infect a final $n \times n$ square with perimeter $4n$.

Hard – Parliament Pacification

Start with any split into two chambers. If someone has at least two enemies in their chamber, move them to the other chamber. They have at most three enemies total, so after moving they have at most one enemy in the new chamber. The total number of same-chamber enemy pairs decreases, so the process terminates. At termination, no member has more than one enemy in their chamber.

Very Hard – Pebble Spreading

The possible values are $n = 1$ and $n = 2$. For $n = 1$, one move clears the first diagonal. For $n = 2$, a short construction clears the first two diagonals. For $n \geq 3$, assign weight 2^{2-i-j} to cell (i, j) . A legal move preserves total weight. Clearing the first three diagonals would force the remaining pebbles into cells whose total possible weight is too small, giving an invariant obstruction.